

Demand Properties: Summary

In what follows u is a *continuous* and *locally nonsatiated* real-valued function representing a preference relation $\succsim$ on the consumption set $X = \mathbb{R}_+^L$. For $(p, w) \gg 0$, define $d(p, w) = \{x \in B(p, w) \mid u(x) \geq u(y) \text{ for every } y \in B(p, w)\}$, where $B(p, w) = \{x \in \mathbb{R}_+^L \mid p \cdot x \leq w\}$.

Let $\bar{U} = \text{Range}(u) \cap (u(0), \infty)$. For $(p, \bar{u}) \in \mathbb{R}_{++}^L \times \bar{U}$, define

$$e(p, \bar{u}) = \min_{x \in \mathbb{R}_+^L} p \cdot x \quad (1)$$

subject to the constraint that $u(x) \geq \bar{u}$. Clearly, for each $p \gg 0$, $e(p, \cdot)$ is strictly increasing on $\bar{U}$. (Proof?)

Theorem 1. (*Properties of $e(\cdot, \bar{u})$.*) For every $\bar{u} \in \bar{U}$, the function $e(\cdot, \bar{u})$ on $\mathbb{R}_{++}^L$ is

- nondecreasing;
- homogeneous of degree 1; and
- concave. □

Let $h(p, \bar{u})$ be the set of solutions to (1). If $e(\cdot, \bar{u})$ is C^1 , then $\partial e(p, \bar{u}) / \partial p_i = h_i(p, \bar{u})$ (Shepard-McKenzie Lemma)¹, implying $h(\cdot, \bar{u})$ is single-valued. The next result follows from Theorem 1 and the Shepard-McKenzie Lemma.

Corollary 1. (*Properties of $h(\cdot, \bar{u})$.*) For every $\bar{u} \in \bar{U}$

- $h(\cdot, \bar{u})$ is homogeneous of degree 0 on $\mathbb{R}_{++}^L$;
- If $e(\cdot, \bar{u})$ is C^2 (implying $h(p, \bar{u})$ is single-valued and C^1), then the matrix

$$\begin{bmatrix} \partial h_1 / \partial p_1 & \dots & \partial h_1 / \partial p_L \\ \vdots & & \vdots \\ \partial h_L / \partial p_1 & \dots & \partial h_L / \partial p_L \end{bmatrix}$$

is a symmetric, negative semidefinite matrix for all $p \gg 0$.²

Corollary 2. If $e(\cdot, \bar{u})$ is C^2 for every $\bar{u} \in \bar{U}$, then for every $(p, \bar{u}) \in \mathbb{R}_{++}^L \times \bar{U}$

$$\sum_{j=1}^L \frac{\partial h_i}{\partial p_j} p_j = 0 = \sum_{i=1}^L \frac{\partial h_i}{\partial p_j} p_i$$

with the first equality holding for $i = 1, \dots, L$ and the second for $j = 1, \dots, L$.

The next Lemma follows from continuity and local nonsatiation of u . If demand is single-valued, the proof is even shorter. *Proof:* Exercise.

¹For background on the hyphenation, and for those historically inclined, you might want to read parts of M. Ali Khan and E Schlee “On Lionel McKenzie’s 1957 Intrusion into 20th Century Demand Theory” *Canadian Journal of Economics* (2016) 49: 589 - 636, available on the Canvas Course page.

²Since a concave function is twice (Fréchet) differentiable almost-everywhere, I could instead simply write “For almost all $p \gg 0$ the matrix ... is a symmetric, nsd matrix.”

Lemma 1. For every $(p, \bar{u}) \in \mathbb{R}_{++}^L \times \bar{U}$, we have that $h(p, \bar{u}) = d(p, e(p, \bar{u}))$ and $d(p, w) = h(p, V(p, w))$. Moreover $V(p, e(p, \bar{u})) = \bar{u}$, and $e(p, V(p, w)) = w$.

The next Corollary is the main result; it (finally) summarizes the properties that demands satisfy. It follows from Lemma 1—which implies the Slutsky equation for C^1 demands—and Corollary 1.

Corollary 3. (*Properties of d .*)

- (Budget Balance) $p \cdot x' = w$ for every $x' \in d(p, w)$;
- d is homogeneous of degree 0 on $\mathbb{R}_{++}^{L+1}$; and
- If $e(\cdot, \cdot)$ is C^2 with $\partial e / \partial \bar{u} > 0$,³ then for every $(p, w) \gg 0$ the matrix

$$\begin{bmatrix} \frac{\partial d_1}{\partial p_1} + \frac{\partial d_1}{\partial w} d_1 & \dots & \frac{\partial d_1}{\partial p_L} + \frac{\partial d_1}{\partial w} d_L \\ \vdots & & \vdots \\ \frac{\partial d_L}{\partial p_1} + \frac{\partial d_L}{\partial w} d_1 & \dots & \frac{\partial d_L}{\partial p_L} + \frac{\partial d_L}{\partial w} d_L \end{bmatrix}$$

is a symmetric, negative semidefinite matrix.

□

Corollary 4. If $e(\cdot, \bar{u})$ is C^2 for every $\bar{u} \in \bar{U}$ —so d is C^1 — then for every $(p, w) \gg 0$

$$\sum_{j=1}^L p_j \left[\frac{\partial d_i}{\partial p_j} + \frac{\partial d_i}{\partial w} d_j \right] = 0 = \sum_{i=1}^L p_i \left[\frac{\partial d_i}{\partial p_j} + \frac{\partial d_i}{\partial w} d_j \right] \quad (2)$$

with the first equality holding for $i = 1, \dots, L$ and the second for $j = 1, \dots, L$.

The first equality follows *directly* from homogeneity of degree zero in (p, w) —differentiate $d(\alpha p, \alpha w) = d(p, w)$ with respect to α ; the second follows *directly* from budget balance—differentiate $p \cdot d(p, w) = w$ with respect to p_j and w , then combine. (Corollary 4 also follows from Corollary 2 and the Slutsky equation; but the independent justification in the preceding sentence is relevant for ASMT IV.)

³By the implicit function theorem it follows that the indirect utility V is C^2 with $\partial V / \partial w > 0$. It follows that Roy's Identity holds and the demands are C^1 .